

Vectors in n Dimensions

ANSWER KEY



Vector addition and scalar multiplication

- 1 Let $\mathbf{u} = \langle 2, -1, 3 \rangle$, $\mathbf{v} = \langle 1, 4, -2 \rangle$. Find $\mathbf{u} + \mathbf{v}$ and $3\mathbf{u} - 2\mathbf{v}$.
 $\mathbf{u} + \mathbf{v} = \langle 3, 3, 1 \rangle$. $3\mathbf{u} = \langle 6, -3, 9 \rangle$, $2\mathbf{v} = \langle 2, 8, -4 \rangle$, so $3\mathbf{u} - 2\mathbf{v} = \langle 4, -11, 13 \rangle$.

Dot product and norm

- 2 For $\mathbf{u} = \langle 1, 2, 2 \rangle$, $\mathbf{v} = \langle 3, -1, 0 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$ and $|\mathbf{u}|$.
 $\mathbf{u} \cdot \mathbf{v} = 1(3) + 2(-1) + 2(0) = 1$. $|\mathbf{u}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$.
- 3 Find the angle between $\mathbf{u} = \langle 1, 0, 0 \rangle$ and $\mathbf{v} = \langle 1, 1, 0 \rangle$ using $\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$.
 $\mathbf{u} \cdot \mathbf{v} = 1$, $|\mathbf{u}| = 1$, $|\mathbf{v}| = \sqrt{2}$. $\cos \theta = \frac{1}{\sqrt{2}}$, so $\theta = 45^\circ$ ($\pi/4$).

The Cauchy–Schwarz Inequality

- 4 State the Cauchy–Schwarz Inequality.
For all vectors $\mathbf{u}, \mathbf{v}$: $|\mathbf{u} \cdot \mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|$.
- 5 Verify the Cauchy–Schwarz Inequality for $\mathbf{u} = \langle 3, 4 \rangle$, $\mathbf{v} = \langle 1, 2 \rangle$.
 $\mathbf{u} \cdot \mathbf{v} = 3 + 8 = 11$. $|\mathbf{u}| = 5$, $|\mathbf{v}| = \sqrt{5}$, so $|\mathbf{u}||\mathbf{v}| = 5\sqrt{5} \approx 11.18$. Since $11 \leq 11.18$, the inequality holds (it is strict here because $\mathbf{u}$ and $\mathbf{v}$ are not parallel).

Unit vectors

- 6 Find the unit vector in the direction of $\mathbf{v} = \langle 3, -4 \rangle$.
 $|\mathbf{v}| = \sqrt{9 + 16} = 5$. Unit vector: $\langle \frac{3}{5}, -\frac{4}{5} \rangle$.

Orthogonality

- 7 Determine whether $\mathbf{u} = \langle 2, -3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2, 4 \rangle$ are orthogonal.
 $\mathbf{u} \cdot \mathbf{v} = 2(1) + (-3)(2) + 1(4) = 2 - 6 + 4 = 0$. Since the dot product is 0, they are orthogonal.
- 8 For $\mathbf{u} = \langle 1, -1, 2 \rangle$, find $|\mathbf{u}|$ and the unit vector $\hat{\mathbf{u}}$.
 $|\mathbf{u}| = \sqrt{1 + 1 + 4} = \sqrt{6}$. $\hat{\mathbf{u}} = \langle \frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \rangle$.